

Literature Status: Negative Answer to Koldobsky's Measure-Slicing Problem

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Status. literature_already_answered (negative; fails already for $k = 1$). This note records an already-known later-literature answer, not a new proof from the run.

Source Question

The source paper is A. Koldobsky, *Slicing inequalities for measures of convex bodies*, arXiv:1412.8550, published in *Advances in Mathematics* 283 (2015), 473–488. In Problem **prob** in the Introduction, Koldobsky asks whether there is an absolute constant C such that, for every n , every $1 \leq k < n$, every origin-symmetric convex body $L \subset \mathbb{R}^n$, and every measure μ with non-negative even continuous density,

$$\mu(L) \leq C^k \max_{H \in Gr_{n-k}} \mu(L \cap H) |L|^{k/n}.$$

The same question is stated in the abstract of the source paper.

Supporting Answer

The supporting paper is B. Klartag and G. V. Livshyts, *The lower bound for Koldobsky's slicing inequality via random rounding*, arXiv:1810.06189. The title and introduction explicitly identify the paper as studying the lower bound for Koldobsky's slicing inequality, and the bibliography cites Koldobsky's 2015 paper.

For a measure μ with continuous density and an origin-symmetric convex body K , Klartag–Livshyts define

$$S_{\mu,K} = \inf_{\xi \in S^{n-1}} \frac{\mu(K)}{|K|^{1/n} \mu^+(K \cap \xi^\perp)}.$$

They recall that Koldobsky proved the general upper bound $S_n \leq \sqrt{n}$. Their Theorem 1 proves the matching lower bound: for every n , there exist a measure μ and a symmetric convex body $L \subset \mathbb{R}^n$ such that for all $\xi \in S^{n-1}$ and all $t \geq 0$,

$$\mu^+(L \cap (\xi^\perp + t\xi)) \leq \frac{C}{\sqrt{n}} \mu(L) |L|^{-1/n},$$

where C is universal.

Identification

Take $t = 0$ in the supporting theorem. Then every central hyperplane section satisfies

$$\mu^+(L \cap \xi^\perp) \leq \frac{C}{\sqrt{n}} \mu(L) |L|^{-1/n}.$$

Equivalently, $S_{\mu,L} \geq c\sqrt{n}$. If the source problem had a dimension-free constant for all k , then the case $k = 1$ would imply

$$\mu(L) \leq C_0 \max_{\xi \in S^{n-1}} \mu^+(L \cap \xi^\perp) |L|^{1/n} \leq \frac{C_0 C}{\sqrt{n}} \mu(L),$$

which is impossible for large n . Thus the proposed absolute-constant inequality in Problem `prob` is false already in codimension one.

The density condition matches the source problem. In the proof, Klartag–Livshyts construct the measure as a convolution of the standard Gaussian measure with an explicitly symmetrized discrete measure, so the resulting density is continuous, non-negative, and even.

Scope

This packet answers only the arbitrary-measure absolute-constant question in arXiv:1412.8550. It does not affect Koldobsky’s positive results for unconditional bodies, for duals of bodies with bounded volume ratio, or for proportional codimension $k \geq \lambda n$. It also does not address Bourgain’s original Lebesgue-volume slicing problem.

For $k = 1$, the later paper gives the sharp order up to constants: Koldobsky’s upper bound gives $S_n \leq \sqrt{n}$, and Klartag–Livshyts give $S_n \geq c\sqrt{n}$.

Evidence

The run indexes were searched for arXiv:1412.8550, the source title, and core terms including `slicing inequalities`, `measures of convex bodies`, `Koldobsky’s slicing inequality`, and `random rounding`. No exact prior packet for arXiv:1412.8550 was found. Related prior slicing packets concern Bourgain’s volume slicing problem and do not duplicate this measure-slicing status.

The source PDF is included as `source_paper.pdf`. The supporting PDF is included as `supporting_paper_1810.06189.pdf`.

References

- [1] A. Koldobsky, *Slicing inequalities for measures of convex bodies*, Advances in Mathematics 283 (2015), 473–488; arXiv:1412.8550.
- [2] B. Klartag and G. V. Livshyts, *The lower bound for Koldobsky’s slicing inequality via random rounding*, arXiv:1810.06189.